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FBE 559: Practice Exam 1

Make sure you sign the sign-in sheet.

Answer all questions.

Keep your answers brief.

For graphs, rough sketches are okay, but label your axes clearly.

Name:

I. WARM-UP

1. Sketch the payoff diagram of a long forward with forward price F on an underlying with price S_T at maturity date T. [5 marks]

2. Suppose that the (annually compounded) one-year interest rate is 100% and it stays constant in time. [10 marks]

   a) What is the price of a one-year bond[1]?

   b) What is the one-year forward price of a one-year bond?

   c) What is the price of a two-year bond?

   d) What is the one-year forward price of a two-year bond?

[1] As ever, a "bond" is a zero-coupon bond with face value one unless stated otherwise.

3. Suppose that the price of cocoa is \$1, the (annually compounded) one-year interest rate is 10%, and it costs \$0.05 a year to store cocoa (paid at the end of the year). What is the one-year forward price of cocoa? [5 marks]

4. How would an increase in the risk of cocoa affect the forward price of cocoa in your answer above? Explain your reasoning very briefly. [5 marks]

5. Suppose that a security X has price 1 today and pays off -10 tomorrow for sure. Can you conclude that there is an arbitrage? If so, what is it? (No proof or explanation is necessary.) [5 marks]

II. SWAPS

There are two dates, Date 0 and Date 1. There are two assets, Asset 1 and Asset 2. Let S_t^n denote the price of Asset n at Date t.

Here, we study a swap, which is similar to a forward, but rather than agreeing to the cash purchase of an asset in the future, you agree to exchange another asset for it. Specifically, here, as in a forward, you exchange nothing today and you get one unit of S^1 at Date 1. Except instead of paying in cash, you deliver f units of of Asset 2.

Here we find the "swap rate" f.

1. Keeping the Date-1 price of Asset 2 S_1^2 and the swap rate f fixed, draw the payoff diagram of the long position in the swap as a function of S_1^1. [5 marks]

2. Write down the Date-1 payoff of the long position swap in terms of S_1^1, S_1^2, and f. [5 marks]

3. Assume no dividends, storage costs etc. Write down a replicating portfolio for the long swap. [5 marks]

4. Maintaining the assumption of no dividends, find an expression for the swap rate f. What result have you used? [5 marks]

5. Suppose that Asset 2 (only) pays dividends. Do you think the swap rate f increases or decreases? Explain briefly. [5 marks]

III. Synthetic Spot Prices

Suppose one-year interest rate is 5%. The one-year forward price of cocoa is \$2.10.

1. Roughly sketch the payoff diagrams of the forward, the underlying, and the bond (at maturity) as a function of the price of cocoa, p_1.

2. Replicate the payoff of cocoa with a portfolio with the forward and bonds. Illustrate the replication graphically.

3. Suppose there is no storage cost/convenience yield. Using your answer above, find the spot price of cocoa p_0.

4. Suppose that the market is 1. If you could trade in actual cocoa, could you construct an arbitrage? If so, how? (You don't have to show it is an arbitrage, just say the strategy.)

IV. THE LAW OF ONE PRICE

Here we consider two derivatives written on gold that can be exercised the first time that the price a unit of gold reaches 3000.

Throughout, denote the price a unit of gold by p_t and the first time $p_t = 3000$ by τ. The continuously compounded interest rate per unit time is constant at r.

Assume (i) that the price will reach \$3000 eventually and (ii) that there are no storage costs or transaction costs of any kind.

1. Consider a derivative, with date-t price denoted by X_t, that gives the right to redeem for *two units* of gold at date τ, the first time that the price of one unit reaches 3000.

   a) What is the price, X_0, of the derivative described above (in terms of other prices and/or the interest rate, etc.)? [5 marks]

   b) Suppose that the price is greater than your answer above. Construct an arbitrage and show explicitly that it is an arbitrage. You can denote the hypothetical price by X_0' and the "correct" no-arbitrage price in your answer above by X_0. [10 marks]

   c) How do your think your answer could change if gold were costly to store? Explain briefly. [5 marks]

2. Now suppose that you have another derivative, with date-t price denoted by Y_t, that gives you the *option* to buy gold for \$2000 at date τ, when $p_t = 3000$ for the first time.

   a) Is the option a type of put, a call, or neither? [5 marks]

   b) What is X_τ, the value of the derivative at date τ, the first time the price is 3000. [5 marks]

   c) Suppose that the interest rate is zero ($r = 0$). What is the price of the option at date 0? [5 marks]

d) Maintain the assumption that $r = 0$. How do you think that the price of the option would change if the volatility of the price of gold increased (and everything else, including the price p_0 of gold today, is unchanged)? Explain your reasoning briefly. [5 marks]

e) Now suppose $r > 0$. How do you think your answer to the previous question would change? I.e. how do you think that the price of the option today would change if the volatility of the price of gold increased? Explain your reasoning briefly. [5 marks]

V. Formulae

$$F = (1+r)p_0 + c,$$

$$F_t(T) = e^{(r-y)(T-t)}p_t,$$

$$S_0 = \frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}\big[S_1\big],$$

$$q = \frac{1+r-d}{u-d},$$

$$\mathbb{E}\big[R_i\big] = R_f + \beta_i\Big(\mathbb{E}\big[R_M\big] - R_f\Big),$$

$$\beta_i = \frac{\mathrm{Cov}[R_i, R_M]}{\mathrm{Var}[R_M]},$$

$$\beta_\theta = \frac{\Delta S_0}{bB_0 + \Delta S_0}\beta_S,$$

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$$\mathrm{Cov}[X,Y] = \mathbb{E}\Big[\big(X - \mathbb{E}[X]\big)\big(Y - \mathbb{E}[Y]\big)\Big] = \mathbb{E}\big[XY\big] - \mathbb{E}\big[X\big]\mathbb{E}\big[Y\big],$$

$$\mathrm{Var}[X] = \mathrm{Cov}[X,X],$$

$$\mathrm{corr}[X,Y] = \frac{\mathrm{Cov}[X,Y]}{\sqrt{\mathrm{Var}[X]\mathrm{Var}[Y]}},$$

$$u = \frac{1}{d} = e^{\sqrt{h}\sigma}\ ,\ q = \frac{e^{hr} - d}{u-d},$$

$$X_t = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}\big[X_T\big],$$

$$c = S\Phi(d_1) - e^{-rT}K\Phi(d_2),$$

$$d_1 = \frac{\log(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}\ ,\ d_2 = d_1 - \sigma\sqrt{T},$$

$$\Delta = \frac{\partial V}{\partial S}\ ,\ \Gamma = \frac{\partial \Delta}{\partial S}\ ,\ \Theta = \frac{\partial V}{\partial t}\ ,\ \nu = \frac{\partial V}{\partial \sigma},$$

$$\phi(z) = \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}\ ,\ \Phi(z) = \int_{-\infty}^{z}\phi(\zeta)d\zeta,$$

$$f(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma}$$